

Junheng Li, Ph.D.

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Professional Experiences

2025/09-present ♦ **Postdoctoral Scholar Research Associate** (AMBER Lab; P.I.: Aaron Ames, Ph.D.)
California Institute of Technology.

Education

2022-2025/08 ♦ **Ph.D., University of Southern California** in Mechanical Engineering.
Advisor: Quan Nguyen, Ph.D.
Thesis: *Hierarchical Optimization-Based Control for Dynamic Loco-Manipulation on Humanoid Robots*

2020-2022 ♦ **M.S., University of Southern California** in Mechanical Engineering.

2016-2020 ♦ **B.S., Loyola Marymount University** in Mechanical Engineering.

Research Experiences

California Institute of Technology (2025/09 - present)

- 2025-2026 ♦ **Model and control guided reinforcement learning for humanoid control** [P1, P2]
- Advisor: Aaron Ames, Ph.D.
 - Exploring principled model-based training-time guidance for learning dynamic and constraint-satisfying locomotion and manipulation.
- ♦ **Accelerating and scaling GPU-based numerical optimization solvers for robotic control** [P2, P3, P4]
- Advisors: Aaron Ames, Ph.D., Ján Drgoňa, Ph.D.
 - Exploring memory-efficient and scalable solvers tailored for parallel computing in training environments, accelerating optimal-control-guided learning for humanoid robots.

University of Southern California (2022/01 - 2025/08)

- 2024 – 2025 ♦ **Survey on Humanoid Loco-manipulation** [J1]
- Advisors: Quan Nguyen, Ph.D., Ye Zhao, Ph.D.
 - Collaborating Institutes: Georgia Tech, CMU, TUM, Duke, Stanford, etc.
 - Reviewed and analyzed the current technical and theoretical progress of humanoid loco-manipulation in control, planning, sensing, learning, and foundation models.
- ♦ **DiffTune and DiffCoTune for Cross-domain Control on Humanoid Robots** [C2, J2]
- Advisors: Quan Nguyen, Ph.D., Naira Hovakimyan, Ph.D., Sheng Cheng, Ph.D.
 - Collaborating Institutes: UIUC, Stanford University.
 - Investigated differentiable-simulator-based optimal control parameter-tuning on humanoid MPC.
 - Investigated differentiable co-tuning of control and system parameters jointly for cross-domain robot control deployment.

- ◇ **Scalable Control of Humanoid Tele-operation through ForceBot** [C3]
 - Advisors: Quan Nguyen, Ph.D., Kaveh A. Hamed, Ph.D., Alexander Leonessa, Ph.D.
 - Collaborating Institutes: Virginia Tech.
 - Implemented long-distance tele-locomotion framework between ForceBot and Hector V2 humanoid with MPC and CoM reference command scaling.
- 2021 – 2025 ◇ **Dynamic Locomotion via Hierarchical Optimizations on HECTOR Humanoid** [P6, P7, J3, C4, C5, C9]
 - Advisor: Quan Nguyen, Ph.D.
 - Developing the hardware, mechatronics, control, and planning for an open-source miniature humanoid robot platform, HECTOR. https://github.com/DRCL-USC/Hector_Simulation
 - Proposed and realized the first-ever continuous dynamic jumping on bipedal robots with hierarchical optimization and variable modeling strategies.
 - Proposed force-and-moment humanoid robot dynamics model and augmented the simplified model with Gait Network for variable-frequency walking control.
- 2023 – 2025 ◇ **Dynamic Humanoid Whole-body Loco-manipulation Control** [C6, C7, P7]
 - Advisor: Quan Nguyen, Ph.D.
 - Investigated kino-dynamic pose optimization for whole-body pose planning in heavy-weight humanoid pushing.
 - Proposed multi-contact external force modeling for controlling humanoid robots perform dynamic object transfer.
- 2022 ◇ **Wheel-legged Robot Navigating High Obstacles via Pose Optimization** [C8]
 - Advisor: Quan Nguyen, Ph.D.
 - Proposed and developed kinematics-based pose optimization framework to optimize the driving pose when traversing high obstacles.

Teaching Experiences

- 2023-2025 ◇ **Teaching Assistant**, University of Southern California.
 - AME 451: Linear Control Systems I (*Fall 2023, Spring 2024, and Spring 2025*)
 - Led discussion sessions, lecture review, and coding sessions.
 - Course covers Laplace transform, frequency domain analysis, compensator design, PID, Root Locus, Bode plots, and Nyquist plot.
- 2024 ◇ **Invited Project Advisor**, University of Southern California.
 - AME 556: Robot Dynamics and Control (*Fall 2024*)
 - Participated in project advising, coding sessions, and project reviews.
 - Course covers robot kinematics, dynamics, optimal control (LQR, QP, MPC, direct optimization), nonlinear control, and reinforcement learning.

Publications

Journal Articles

- J1

 Z. Gu, J. Li, W. Shen, *et al.*, “Humanoid locomotion and manipulation: Current progress and challenges in control, planning, and learning,” *IEEE/ASME Transactions on Mechatronics (to appear)*, 2025.
- J2

 L. Krishna, S. Cheng, J. Li, Q. Chen, N. Hovakimyan, and Q. Nguyen, “Diffcotune: Differentiable co-tuning for enhanced cross-domain robot control,” *IEEE Robotics and Automation Letters (RA-L)*, 2025.

- J3 J. Li and Q. Nguyen, “Dynamic walking of bipedal robots on uneven stepping stones via adaptive-frequency mpc,” *IEEE Control Systems Letters*, vol. 7, pp. 1279–1284, 2023.

Conference Proceedings

- C1 S. A. Esteban, J. Li, V. Kurtz, and A. D. Ames, “On surprising effects of risk-aware domain randomization for contact-rich sampling-based predictive control,” in *Workshop of 2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- C2 Q. Chen*, J. Li*, S. Cheng, N. Hovakimyan, and Q. Nguyen, “Autotuning bipedal locomotion mpc with grfm-net for efficient sim-to-real transfer,” in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2025.
- C3 A.-C. He*, J. Li*, J. Park*, *et al.*, “A novel telelocomotion framework with com estimation for scalable locomotion on humanoid robots,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- C4 J. Li, Z. Duan, J. Ma, and Q. Nguyen, “Gait-net-augmented implicit kino-dynamic mpc for dynamic variable-frequency humanoid locomotion over discrete terrains,” in *Robotics: Science and Systems*, 2025.
- C5 J. Li*, Z. Le*, J. Ma, and Q. Nguyen, “Adapting gait frequency for posture-regulating humanoid push-recovery via hierarchical model predictive control,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- C6 J. Li and Q. Nguyen, “Kinodynamic pose optimization for humanoid loco-manipulation,” in *2023 IEEE-RAS 22nd International Conference on Humanoid Robots (Humanoids)*, 2023, pp. 1–8.
- C7 J. Li and Q. Nguyen, “Multi-contact mpc for dynamic loco-manipulation on humanoid robots,” in *2023 American Control Conference (ACC)*, 2023.
- C8 J. Li, J. Ma, and Q. Nguyen, “Balancing control and pose optimization for wheel-legged robots navigating high obstacles,” in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2022, pp. 8835–8841.
- C9 J. Li and Q. Nguyen, “Force-and-moment-based model predictive control for achieving highly dynamic locomotion on bipedal robots,” in *2021 60th IEEE Conference on Decision and Control (CDC)*, IEEE, 2021, pp. 1024–1030.

Preprint

- P1 M. Dai, W. D. Compton, J. Li, L. Yang, and A. D. Ames, *Walk the plank: Physics-guided rl for agile humanoid locomotion on constrained footholds*, 2026.
- P2 J. Li, L. Wu, S. A. Esteban, L. Yang, J. Drgoňa, and A. D. Ames, *Accelerating and scaling mpc-guided reinforcement learning for humanoid locomotion and manipulation*, 2026.
- P3 J. Li, L. Yang, and A. D. Ames, *Mirror: Visual motion imitation via real-time retargeting and teleoperation with parallel differential inverse kinematics*, 2026.
- P4 L. Wu, B. Yang, J. Li, *et al.*, *π Mpc: A parallel-in-horizon and construction-free nmpc solver*, 2026.
- P5 L. Yang, J. Li, N. Poddar, *et al.*, *Handoff: Humanoid agentic task-space whole-body control via distilled complementary teachers*, 2026.
- P6 J. Li, O. Kolt, and Q. Nguyen, *Continuous dynamic bipedal jumping via adaptive-model optimization*, 2024.
- P7 J. Li, J. Ma, O. Kolt, M. Shah, and Q. Nguyen, *Dynamic loco-manipulation on hector: Humanoid for enhanced control and open-source research*, 2023.

Invention Patents

- 11 Q. Nguyen, J. Li, and J. Ma, "A bipedal robot for dynamic and robust locomotion in challenging environments," WO2025049602, March 6th, 2025.
- 12 Q. Nguyen, J. Li, and J. Ma, "Design and control of wheel-legged robots navigating high obstacles," WO2023205766A1, October 26th, 2023.

Services

- Organizing Committee*
 - ◇ Registration Technical Lead of Robotics: Science and Systems (RSS) 2025
- Reviewer*
 - ◇ International Journal of Robotics Research (IJRR), IEEE Robotics and Automation Letters (RA-L), IEEE/ASME Transactions on Mechatronics (TMECH), IEEE Transactions on Automation Science and Engineering (T-ASE), IEEE/CAA Journal of Automatica Sinica, Journal of Field Robotics (JFR), Autonomous Robots (AURO), IEEE Control System Letters (L-CSS), Robotics and Autonomous Systems (RAS), Robotica, Advanced Robotics Research, Frontiers in Mechanical Engineering, International Journal of Robotics and Automation
 - ◇ IEEE International Conference on Robotics and Automation (ICRA), IEEE International Conference on Intelligent Robots and Systems (IROS), IEEE International Conference Humanoid Robots (Humanoids), IEEE Conference on Decision and Control (CDC), Robotics: Science and Systems (RSS), Conference on Robot Learning (CoRL)
- Outreach*
 - ◇ *Coordinator of Lab Outreach* for Dynamic Robotics and Control Lab, hosted more than 50 lab tours and outreach events.

Invited Talks, Presentations, and Demonstrations

- 2026/03 ◇ *"Toward Deployable Humanoid Robots"*
Invited talk and panel, Caltech Entrepreneur Forum
- 2026/01 ◇ *"Toward Model-guided Learning on Humanoid Robots"*
Invited talk, Georgia Tech LIDAR Group
- 2025/11 ◇ *"Towards Model-guided Learning on Humanoid Robots"*
Invited talk, Loyola Marymount University
- 2025/04 ◇ *"Hierarchical Optimization-based Control for Dynamic Loco-manipulation on Humanoid Robots"*
Invited talk, Caltech AMBER Lab
- 2024/11 ◇ *"HECTOR: A Novel Mini-humanoid Platform for Accessible and Accelerated Research and Development"*
Invited talk, USC Robotics Seminar
- 2024/06 ◇ *"Demonstration of Dynamic Loco-manipulation on HECTOR: Humanoid for Enhanced Control and Open-source Research"*
Demonstration, IEEE International Conference on Robotics and Automation (ICRA)
- 2023/12 ◇ *"Kinodynamic Pose Optimization for Humanoid Loco-manipulation"*
Poster, IEEE International Conference on Humanoid Robots (Humanoids)
- 2023/11 ◇ *"Toward Dynamic Locomotion and Loco-manipulation on Humanoid Robots via Model Predictive Control with Linear Dynamics Models"*
Invited talk, LMU Department of Mechanical Engineering
- 2023/10 ◇ *"Toward Dynamic Locomotion and Loco-manipulation on Humanoid Robots via Model Predictive Control with Linear Dynamics Models"*
Invited talk, 42nd Southern California Control Workshop

- ◇ “*Demonstration of Dynamic Locomotion on Bipedal Robots via Force-and-moment-based Model Predictive Control*”
Demonstration, IEEE International Conference on Intelligent Robots and Systems (IROS)
- 2023/05 ◇ “*Multi-contact Mpc for Dynamic Loco-manipulation on Humanoid Robots*”
Oral Presentation, American Control Conference (ACC)
- ◇ “*Dynamic Walking of Bipedal Robots on Uneven Stepping Stones via Adaptive-frequency MPC*”
Oral Presentation, American Control Conference (ACC)
- 2023/02 ◇ “*Toward Dynamic Locomotion and Loco-manipulation on Humanoid Robots via Model Predictive Control*”
Invited talk, USC Robotics Seminar
- 2022/10 ◇ “*Balancing Control and Pose Optimization for Wheel-legged Robots Navigating High Obstacles*”
Oral Presentation, IEEE International Conference on Intelligent Robots and Systems (IROS)
- 2021/11 ◇ “*Force-and-moment-based Model Predictive Control for Achieving Highly Dynamic Locomotion on Bipedal Robots*”
Oral Presentation, IEEE Conference on Decision and Control (CDC)

Mentoring Experience

Graduate Students:

- 2025 ◇ Lizhi Yang, Ph.D. in Mechanical Engineering, Caltech
- ◇ Sergio Esteban, Ph.D. in Mechanical Engineering, Caltech
- 2024-2025 ◇ Dakota Mercer, M.S. in Mechanical Engineering, USC
Project: *Hierarchical-Optimization for Dynamic Stair Climbing on Bipedal Robots*
- ◇ Ziwei Duan, M.S. in Computer Science, USC
Project: *Gait-Net: A Data-driven Approach to Enhance Bipedal Locomotion Control*
- 2023-2024 ◇ Zhanhao Le, M.S. in Electrical Engineering, USC
Project: *Bipedal Robot Push Recovery via Hierarchical-MPC*
Now: Prospective Ph.D. student
- ◇ Omar Kolt, M.S. in Mechanical Engineering, USC.
Project: *Control and Software Infrastructure of HECTOR Humanoid*
Now: Software Engineer at Tesla, Optimus Team
- ◇ Omar Berra, M.S. in Mechanical Engineering, USC
Project: *Design and Whole-body Control of Bipedal Wheel-legged Robot*
Now: Testing Engineer at the Boring Company
- 2022-2023 ◇ Han Gong, M.S. in Mechanical Engineering, USC
Project: *Terrain-aware Bipedal Robot Control and Simulation Design*
Now: Ph.D. student at UMass Amherst
- 2021-2022 ◇ Xinyu Zhu, M.S. in Mechanical Engineering, USC
Project: *Uneven Terrain Locomotion of Wheel-legged Robots via Whole-body Control*
Now: Engineer at Xiaomi Robotics Lab
- ◇ Tiansheng Wu, M.S. in Mechanical Engineering, USC
Project: *Terrain-aware Trajectory Optimization and Control on Bipedal Robots*

Undergraduate Students:

- 2024-2025 ◇ Rodrigo Andrade, B.S. in Mechanical Engineering, USC
Project: *Design-control Co-optimization of Mini Humanoid Robot*

- ◇ Nathan Chun, B.S. in Mechanical Engineering, USC
Project: *Humanoid Loco-manipulation and Contact-implicit MPC*
- ◇ Bill Guo, B.S. in Electrical Engineering, USC
Project: *Electronic System Design of Humanoid Robot*
- 2022-2024 ◇ Mana Shah, B.S. in Mechanical Engineering, USC
Project: *Design of a Mini Humanoid Robot Platform*

High School Students:

- Summer 2024 ◇ Chinmay Ramamurthy, Ethan Le, Ian Chen, Perceiver Summer Research
Project: *Modeling and Control of Series Elastic Actuators for Knee Exoskeletons*
- ◇ Jonathan Li, Ted Han, Sophia Fu, Perceiver Summer Research
Project: *Package Transferring with UAV and Passive Manipulation Mechanism*
- Summer 2022 ◇ Dylan Dharwadkar, USC SHINE Program
Project: *Swing Trajectory Design and Control in Bipedal Robot Walking*

Media and Press

- HECTOR Humanoid ◇ IEEE Spectrum: [Continuous Dynamic Jumping on HECTOR](#)
- ◇ IEEE RAM: [The Next Generation of Robotics](#)
- ◇ USC Today: [HECTOR at the Ginsburg Hall's Grand Opening Ceremony](#)
- Wheel-legged Robot ◇ USC News: [Navigate Terrain and Combat Obstacles](#)
- ◇ Tech Briefs: [Wheel-Legged Robots Navigating High Obstacles](#)
- Service ◇ IEEE TelePresence: [RoboPalooza Event at Peterman Hill in Lucerne Valley](#)
- ◇ USC Viterbi News: [Robotics Open House 2024](#)